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Cryopreservation of NK and T cells without DMSO for adoptive cell-based immunotherapy

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Abstract

Dimethylsulfoxide (DMSO) being universally used as a cryoprotectant in clinical adoptive cell therapy settings to treat hematological malignancies and solid tumors is a growing concern, largely due to its broad toxicities. Its use has been associated with significant clinical side effects—cardiovascular, neurological, gastrointestinal and allergic—in patients receiving infusions of cell therapy products. DMSO has also been associated with altered expression of natural killer (NK) and T cell markers and their *in vivo* function, not to mention difficulties in scaling up DMSO-based cryoprotectants which introduce manufacturing challenges for autologous and allogeneic cellular therapies, including chimeric antigen receptor (CAR)-T and CAR-NK cell therapies. Interest in developing alternatives to DMSO has resulted in the evaluation of a variety of sugars, proteins, polymers, amino acids and other small molecules and osmolytes as well as modalities to efficiently enable cellular uptake of these cryoprotectants. However, the DMSO-free cryopreservation of NK and T cells remains difficult. They represent heterogeneous cell populations that are sensitive to freezing and thawing. As a result, clinical use of cryopreserved cell therapy products has not moved past the use of DMSO. Here, we present the state-of-the-art in the development and use of cryopreservation options that do not contain DMSO toward clinical solutions to enable the global deployment of safer adoptively-transferred cell-based therapies.

1. Introduction

The adoptive transfer of *ex vivo* expanded, engineered or native immune [T and natural killer (NK)] cells to create enhanced effector cells capable of highly specific cancer targeting has resulted in remarkable clinical responses against previously incurable tumors[1–4]. In the context of adoptive transfer of these cells into human subjects, cryopreservation has become important to enable a continuous quality-controlled supply of cell therapy products[5]. Apart from being critical to the supply of cells for therapy, cryopreservation

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also bridges the gap between manufacturing and clinical administration, enabling centralized manufacturing models[6].

During the past fifty years, the field of cryobiology has seen tremendous advances and breakthroughs. Much of the early work focused on understanding how cells are affected by ice nucleation and ice growth during freezing and/or warming [7]. More recently, other work has demonstrated there are biochemical effects associated with cryoinjury [8]. In order to mitigate the negative effects of cryoinjury, cryoprotectants (cryoprotective agents or CPAs) are used, which are employed to preserve cells by controlling the rates of water transport during freezing, nucleation, and ice crystal growth.

Despite extensive research and an improved understanding of the molecular and biophysical mechanisms behind the freezing and thawing processes, however, cryopreservation science has converged around the use of DMSO as the main cryoprotectant. The use of DMSO is widespread because it is very efficient at protecting the cells intracellularly during cryopreservation [9]. Despite its efficiency in protecting cells from cryodamage, DMSO has been associated with significant clinical toxicities [10], triggering intense study into alternatives.

Most of the CPA formulations tested on NK and T cells have so far relied on DMSO. Not only do these cells represent functionally heterogeneous cell types that are sensitive to cryopreservation, but DMSO has been reported to cause detrimental alterations to their function [11,12]. For that reason, safe, infusible DMSO-free solutions that can be manufactured at scale under GMP conditions specifically tailored for these cells are needed.

Here, we review the current state-of-the-art in the cryopreservation of T and NK cells without DMSO, with a particular focus on new formulations and approaches which have shown to result in the recovery of viable effector functions of these cells post-thaw.

2. Mechanisms of cellular damage during cryopreservation

2.1 Cell damage in response to freezing and thawing

During cryopreservation and thawing, mammalian cells undergo a series of chemical, mechanical and thermal stresses that cause physical damage and can result in cell apoptosis or necrosis [9]. Cellular damage during cryopreservation can occur in response to multiple factors, including intracellular ice due to cooling rate, osmotic damage [13], and cryoprotectant (CPA) toxicity [14]. Cryoinjury occurs in response to a drop in temperature during which freezing occurs and ice crystals form, and is directly impacted by the rate of cooling during the freezing process. Intracellular water content is one of the main factors driving the extent of cryoinjury and, as a result, cell recovery post thaw. Based on insights into cellular damage during freezing, Mazur and colleagues proposed a “two-factor hypothesis” which relates the rate of cooling to cellular damage [15]. When cooling rates are too rapid for water to leave the cell, intracellular ice crystal formation tends to occur. Intracellular ice causes lesions at the cell or plasma membrane and, ultimately, cell damage and even death [16,17]. Conversely, at cooling rates that are too slow, ice will form in the extracellular environment, and water dehydration will occur due to imbalance in osmolality

between intracellular and extracellular environments. When water dehydration is too severe due to exposure of the cell to freezing-concentrated extracellular solutes and extracellular ice, cell shrinkage results in damage and death. This is called the “solution effect.”

Osmotic imbalance may be an additional issue with too rapid cooling rates, combined with suboptimal storage and thawing rates that are too slow. An optimal cooling rate, which avoids ice crystal-induced damage and solution effects, depends on the cell properties, as well as the cells’ permeability to water and CPAs and should be optimized for each cell type. Generally, cells with highly permeable membranes require faster cooling rates. For instance, red blood cells have highly permeable membranes, and require optimal cooling rates of approximately $-2,500\text{ K/min}$ [18]. On the other hand, cells with less permeable membranes, such as T or NK cells, need to be cooled at much slower rates ($\sim -1^\circ\text{C/min}$). Interactions between the cell and extracellular ice which forms during cooling also contributes to cell damage. However, this is typically less damaging than in the case of too fast cooling rates which induce intracellular ice.

Thawing rate also affects cell injury, but this is dependent on prior cooling rate. Generally, in scenarios of slow cooling, fast thawing can avoid recrystallization during rewarming. Conversely, while slow thawing is assumed to allow for osmotic re-equilibration, long exposure times to CPAs can be detrimental to cells. As a result, a limited number of studies exists on slow thawing cell therapy products [19]. Conversely, it has been suggested that with fast cooling (non-vitrified), water that has not crystallized could recrystallize during thawing, particularly at slow thawing rates [20]. In the case of vitrification, rapid thawing was shown to avoid ice recrystallization compared to samples that were thawed slowly [21].

While cell damage during rewarming has been considered a critical factor affecting cell viability after cryopreservation [22], recent data have indicated that cooling rate drives the effect of rate of thawing on immune cell recovery [20]. When the rate of cooling is slow (-1°C/min or slower), the rate of thawing was shown to not affect recovery of viable immune cells. At rapid cooling rates (-10°C/min), on the other hand, a reduction in recovered viable cell number was observed, though only with slow rates of thawing ($1.6\text{ }^\circ\text{C min}^{-1}$ and $6.2\text{ }^\circ\text{C min}^{-1}$) [20]. With regard to cooling and thawing processes, ice recrystallization has been cited as a major mechanism of damage due to intracellular ice [23], while osmotic shock has been recognized as causing cellular damage during thawing, particular with rapid cooling rates [24]. Rapid thawing is generally able to avoid ice recrystallization, thus reducing cellular damage, and can improve survival of samples that have been cooled rapidly. Apart from slow cooling (for example, at a typical rate of -1°C/min), cells are also cryopreserved by vitrification, which uses cooling rates of above $-15,000^\circ\text{C/min}$ [25]. Vitrification can be described as very rapid cooling wherein water does not have time to form ice crystals, but rather solidifies into a glass-like state. As a result, no ice forms. Critical cooling and warming rates are important in vitrification, as they ensure that no ice is formed during the cooling and thawing steps [26]. To achieve vitrification, high concentrations of CPAs ($>4\text{ M}$) and high cooling rates ($>-15,000^\circ\text{C/min}$) are typically needed. Though no ice is formed, toxicity due to high CPA concentrations is a risk with vitrification. Moreover, like slow cooling, vitrification requires cell-specific optimization.

Related to the cooling rate is the nucleation temperature, which also affects cell survival [27]. Ice nucleation, the process of liquid water freezing into solid ice, is influenced by the degree supercooling of a sample—the drop in temperature below its freezing point without freezing [28]. Supercooling, however, typically occurs as a largely uncontrolled phenomenon due to the removal of latent heat that can induce variability in cellular responses to cryopreservation [29]. Ice nucleation and supercooling have been discussed more extensively elsewhere [27,30]. While it has been reported that cells can tolerate being supercooled by about 2 to 10°C before ice crystals form [29], this is a variable process that depends on cell properties. Additional mechanical stresses the cells experience include those caused by detachment from the culture plate, dissociation into single cells, and those caused by centrifugation as a common processing step during freezing, thawing or CPA removal.

2.2 CPAs reduce freezing-induced cryodamage

CPAs are compounds which are used to mitigate the damaging effects of the freezing process [7]. Ideal cryoprotectants should be non-toxic to cells and should not precipitate in aqueous environments [31]. During freezing, CPAs can replace as much as half of the water inside cells [32]. In vitrification, CPA concentrations are typically very high (~ 4–8 M) [33], required to inhibit ice crystal formation, unlike in slow cooling in somatic cell applications, which uses low concentrations of cryoprotectants, typically under 1M.

Two main categories of CPAs exist: cell-penetrating and non-penetrating. CPAs have particular value when they are able to cross the cell membrane and replace water loss to avoid ice crystal formation. These are termed cell-penetrating CPAs, the most common of which is DMSO. Generally, they act by interacting with water largely through hydrogen bonding, reduce the amount of water available to form ice, and lower its melting point, thus protecting the cells from freezing-induced damage. In addition, they maintain intra- and extracellular solute concentrations and, in turn, reduce ice formation. CPAs which are unable to cross the cell membrane and penetrate the cells are termed non-cell-penetrating, or simply non-penetrating, CPAs. They can be osmotically active (such as disaccharides), or osmotically inactive (such as proteins or polymers). Osmotically active CPAs act by promoting cellular dehydration since they increase osmolality of the medium, thereby reducing intracellular ice. Osmotically inactive CPAs act by stabilizing cell membranes [34,35] or by absorbing water molecules to keep them thermally inert in a glassy state [36].

Though their mechanisms of action vary, the presence of CPAs on both sides of the membrane can improve cryoprotection through various mechanisms which include osmotic stabilization and reduction of intracellular ice crystal formation, as mentioned above. The benefit of cell-penetrating CPAs to cell viability post-thaw is well known [37]. These CPAs will inevitably be present on both sides of the membrane, and will partition until reaching a concentration equilibrium. The extracellular CPA compartment will thus contribute to maintaining an unfrozen extracellular fraction [38]. In many cases, benefiting from the intracellular presence of CPAs during cryopreservation is desirable.

Generally, for a given CPA, lower concentrations require high cooling rates and higher warming rates in order to avoid ice formation [39]. For vitrification, cooling rates must be higher than the critical cooling rate (CCR)—the minimum rate to prevent ice formation.

Studies on the relationship between critical cooling rate and CPA concentration have shown that, as CPA concentration increases, the CCR decreases [39,40]. For glycerol-water solutions, for instance, an exponential relationship has been shown to exist between concentration (16–56% w/v) and cooling rate [41], with higher concentrations requiring lower cooling rates. While some studies have shown that cooling rates could not allow for separation of lymphocyte populations solely on the basis of susceptibility to freezing [42], recent data have shown that cooling rate can affect immune cell recovery, but only when suboptimal thawing rates are used, largely due to changes in ice crystal structure, as we discussed above [20].

3. Known issues with DMSO as a cryoprotectant

3.1 Toxicities of DMSO and effects on cell function

The vast majority of current clinical and pre-clinical cryopreservation protocols uses DMSO at concentrations of 10% [43] in combination with xenogeneic component such as fetal bovine serum (FBS), or serum albumin. Safety concerns associated with DMSO have long been documented [44]. DMSO is undesirable because it carries the risk of considerable toxicities to patients [45,46], while the suitability of its use in cellular therapy products which will be injected in a localized manner into tissues such as the brain is being questioned [38]. In addition, its use has been associated with significant clinical issues in patients receiving infusions of cryopreserved cell therapy products, including allergic, gastrointestinal, cardiac and neurological complications, including seizures [47–54]. *In vitro*, the presence of DMSO has been proven to cause cell death [55] and compromise cell permeability [56]. It has, moreover, been reported as causing detrimental issues to the function of immune cells, including a suppression of pro-inflammatory cytokine and chemokine production, and a loss of immune cell viability [57]. More specifically, the presence of DMSO in cryopreservation media was shown to alter physiological features of multiple cell types [58,59] and induce unwanted epigenetic changes [60,61]. In the context of CAR-T cell infusions, DMSO has been observed to cause nausea, vomiting and hypotension [62], as well as anaphylaxis and severe infusion reactions [63]. DMSO was also reported as having a pathogenic role on NK cell activity through uncontrolled *in vivo* activation of these cells accompanied by interferon- γ (IFN- γ) and granzyme B production in acetaminophen-induced liver injury [64].

3.2 Cell therapy manufacturing issues due to DMSO

From a manufacturability perspective, DMSO poses problems when it comes to scale up, with regulatory issues being a concern as well. Safety concerns with manufacturing processes using DMSO as a solvent have been raised [44,65]. It is considered not compatible with sterile weldable polysulfone, polycarbonate and flexible PVC tubing, thus posing problems in terms of formulation compatibility [66]. DMSO is also able to react rapidly and exothermically with a variety of materials, and is able to rapidly penetrate the skin (176 ± 42 g/m²/hr). The FDA has, as a result of reported toxicity concerns with DMSO, been wary of extending therapeutic approvals of its use. Recent studies have demonstrated that DMSO is not inert and that its use in cryopreservation should be reconsidered [67]. The manufacturing of autologous and—especially—allogeneic cellular therapies that are directly

infusible requires controllable and reproducible cryopreservation processes that result in biocompatible and safe solutions and which can be manufactured at scale. Currently, despite many attempts and multiple DMSO-free products on the market, there is no viable solution for the efficient cryopreservation of T or NK cells—this includes engineered variants—without DMSO.

3.3 Removal of DMSO prior to infusion

If cells are cryopreserved with DMSO, a washing step is often introduced prior to infusion of the cellular product into recipients, in order to remove or dilute the DMSO. This is done for two main reasons: to limit the *in vivo* toxicity-related side effects to treated subjects after infusion, and to protect the cellular product from DMSO-related toxicities. The washing step is performed at the point-of-care facility after the cells are thawed. The requirement and extent of a washing procedure or post-thaw manipulation may be limited by regulatory requirements or the availability of cell processing equipment at the point-of-care facility. Though effective at reducing the concentration of DMSO in the cell suspension, washing also causes loss of viable cells, diminishing the yield and potency of the cell therapy product. It also carries risks of damaging the cells due to the processing involved. Washing is usually performed in a gradual, stepwise dilution manner. This is done so as to avoid osmotic damage to the cells that can occur during CPA removal. The washing solution often contains non-permeating CPAs, such as sugars, to moderate the penetration of water into the cell, which tends to occur at a faster rate than the rate of diffusion of the CPA out of the cell, and thereby avoid osmotic swelling and cellular damage as a result [68]. Optimization of CPA removal requires knowledge of the osmotic tolerance limits of the particular cell type being thawed, and a combination of parameters including washing solution composition and addition rate. For solutions which cause higher osmotic differences between the inside and outside of the cells, dilution rates tend to be slower. Unlike intracellular cryoprotectants, extracellular cryoprotectants don't have the requirement of having to diffuse out of the cell, facilitating their removal. Mathematical models have been developed to design optimal CPA addition and removal processes [69,70]. The addition of betaine to solutions of disaccharides or amino acids was recently shown to promote efficient one-step CPA removal from thawed red blood cell suspensions [71].

Generally, DMSO is removed either manually or using devices. Device-assisted removal of CPAs enables the automation of the removal process and operates on the basis of sequential dilution and centrifugation or dilution and filtration steps [72]. More recent systems have included the development of microfluidic devices incorporating dead-end filtration capabilities to remove DMSO and protein impurities from the thawed cell suspension. Using such a system, over 85% DMSO could be removed from suspensions of thawed fibroblasts and mesenchymal stem cells, with cell recoveries of 73% or higher[73]. However, this solution would currently only be viable for small volume samples of high cell density.

4. Alternatives to DMSO

4.1 Sugars, sugar alcohols, polymers and synthetic proteins

A significant amount of research has also been done on alternatives to DMSO. Small molecules, carbohydrates, amino acid derivatives and analogues, have all been studied for their cryoprotective properties. More recently, some of the focus has been on the development of ice recrystallization inhibitors, which can mitigate thawing-related cryodamage[74]. Most of the innovation to date has relied on replacing DMSO with small molecule, sugar or protein-based alternatives. Early strategies for cryopreservation utilized glycerol [75–77] or glucose, which provides an energy source [78,79]. However, at the high concentrations used glycerol can be toxic and has to be removed prior to cell infusion [47,80,81], while glucose has been known to carry manufacturability issues [78]. To improve performance, other sugars including sucrose [82–84] and dextran[85], as well as molecules such as ethylene glycol [86], propylene glycol [87], ectoine [88–90], proline [91,92], mannitol [93,94], hydroxyethyl starch [36,95,96] and trehalose [97–99] as well as amino acids and derivatives including poly-L-lysine [100,101], glycine [102–104] and isoleucine [105,106] have been studied, alongside polymers such as Poloxamer188 [107,108] with human serum albumin [109] or Plasma-Lyte [110]. Using various osmolytes, Pi and colleagues optimized a cryopreservation mixture composed of sugars, sugar alcohols and amino acids as DMSO-free cryoprotectants for both Jurkat cells, an immortalized T cell line, and T cell subsets isolated from healthy donor peripheral blood mononuclear cells (PBMCs). Optimization of cryoprotectant composition with Jurkat cells showed the highest recoveries (>80%) with formulations of sucrose-glycerol-isoleucine, trehalose-glycerol-isoleucine and maltose-glycerol-isoleucine [111]. When these formulations were used to cryopreserve various subpopulations of T cells, the authors noted that each cell type responded differently to cryopreservation and that no single CPA formulation could be used to equally cryopreserve every cell type.

The addition of rho kinase (ROCK) inhibitors [112] has also been evaluated to enhance the performance of non-DMSO molecules. These are compounds that target rho kinase and inhibit the ROCK pathway. Elevated ROCK signaling activity has been shown to contribute to cell apoptosis, prompting the use of inhibitors to provide pro-survival responses [113]. In that context, the use of ROCK inhibitors in cryopreservation of cells has been evaluated for their ability to contribute to cell survival [114].

Synthetic antifreeze (glyco)protein mimics have been developed as alternatives to natural antifreeze glycoproteins[115]. These mimetics, which have included peptides and glycoproteins, have shown to possess potent ice recrystallization inhibition activity. They have included *O*-aryl-glycosides[116], N-aryl-d-alidonamides[117], poly(vinyl alcohol) [115,118], polyampholytes[119,120], and polyproline[121,122]. The toxicities of various cryoprotectants were surveyed recently.

Encapsulation of cells inside covalently-crosslinked hydrogels is another strategy that has been developed to protect cells during cryopreservation[123]. Hydrogel-based cell encapsulation technology aims to provide a 3D microenvironment surrounding the encapsulated cells by mimicking the extracellular matrix, thus minimizing cryopreservation-

induced mechanical and osmotic injury, and buffering damaging ice crystal formation. Cao and colleagues developed an alginate hydrogel microfiber approach to cryopreserve human red blood cells in the presence of very low concentrations (2.5–5% v/v) of glycerol as the CPA. Glycerol could, moreover, be removed by washing with a sodium chloride solution [124]. Khetan and colleagues investigated the effect on human mesenchymal stem cell viability and post-thaw recovery of cryopreservation using hyaluronic acid and polyethylene glycol-crosslinked hydrogels, deployed via a gradual cooling and freezing protocol with or without DMSO. They showed that hydrogel encapsulation not only preserved the cell viability, but also maintained stem cell differentiation potential after cryopreservation only in the presence of DMSO [125]. In the absence of DMSO, cell viability was only ~0%–5%. Elsewhere, Chen and colleagues developed a multi-component approach to cryopreserve islet β cells by combining microfluidic encapsulation of the cells inside crosslinked hydrogels and nano-delivery of trehalose as the sole CPA. After intracellular delivery of trehalose by nanoparticles, β cells were encapsulated inside calcium alginate hydrogels and cryopreserved by slow freezing. This resulted in post-thaw viability and functionality that were comparable to those of fresh cells. The cryopreserved β cell-laden hydrogels could be directly transplanted to diabetic rats, where they were shown to regulate blood glucose comparably to fresh cells [126].

4.2 Enhancing intracellular delivery of CPAs

In general, non-DMSO CPAs fall in one of two categories: non-cell-penetrating and cell-penetrating CPAs. A list of CPAs that have been evaluated in the context of cryopreservation as potential replacements to DMSO are listed in Table 1. Intracellular delivery of CPAs is desirable to provide the cell with protection from lethal intracellular processes during freezing, as discussed earlier (Section 2.1), such as by inhibiting intracellular ice formation. For those cryoprotectants which are unable to spontaneously diffuse into the cells, their efficient intracellular delivery is a major hurdle [127]. This is because at temperatures below zero, CPA permeability is reduced, requiring their addition at positive temperatures which is sometimes done even at ambient temperatures.. However, the poor uptake of many osmolytes was shown to require long incubation times.[128] Moreover, the permeability of immune cells to CPAs and water is reduced compared to many other cell types (for example, water permeability of 0.188 $\mu\text{m}/\text{min}/\text{atm}$ for human T cells compared to 2.2 $\mu\text{m}/\text{min}/\text{atm}$ for megakaryocytes) [129,130].

To address some of these challenges, the need for tight control of CPA loading and unloading has been studied within the context of vitrification[70] and droplet microfluidics [131–133]. Approaches at enhancing CPA penetration have included electroporation [134], the use of transmembrane pores including macrocycles [135] and liquid-phase endocytosis [136]. The overexpression of aquaporins has also been studied [137,138]. Alongside these, nanoparticle-mediated intracellular delivery of non-permeable CPAs is an emerging approach to cryopreservation that has been gaining interest. This approach does not introduce physical or chemical modifications to the cells, instead relying on endocytosis-mediated internalization to deliver cryoprotectants, and typically employs the same CPAs used in solution-based cryopreservation procedures. So far, nanoparticles have been mostly used to deliver the cryoprotectant trehalose. Rao *et al.* synthesized

pH responsive genipin-crosslinked Pluronic F127-chitosan nanoparticles to encapsulate trehalose. Its intracellular delivery protected primary human adipose-derived stem cells from cryoinjury following cryopreservation and significantly rescued loss of cell viability [139]. In another study, a cold-responsive polymer (poly (Nisopropylacrylamide-co-butyl acrylate)) was utilized to synthesize nanoparticles for the intracellular delivery of trehalose. The rapid cold-triggered intracellular release of trehalose contributed to the retention of post-thaw viability of mammalian cells that was comparable to that using DMSO [140]. Our group developed chitosan-sodium tripolyphosphate (TPP) nanoparticles to deliver trehalose into NK cells (NK-92). After freezing NK cells in the presence of the trehalose-encapsulated nanoparticles, post-thaw recovered cells exhibited comparable viability and higher cytotoxicity against target tumor cells compared to NK cells frozen using 10% DMSO-containing medium[141].

As a cryopreservation strategy, intracellular delivery of CPAs to immune cells requires considerations on the fate—including the degradation rate—of the CPA once inside the cell, as well as on any potential changes to the behavior or metabolic function of immune cells that are bearing such cargo. To date, studies have mostly focused on investigating the effects of intracellular delivery of CPAs to immune cell effector function, while deeper insights into the underlying molecular and functional changes require further investigation.

4.3 Additional challenges with DMSO-free cryopreservation approaches

Though promising insofar as their ability to reduce or eliminate the use of DMSO, DMSO-free cryopreservation formulations have failed to so far demonstrate universal applicability across a wide range of cell types, with studies showing that these strategies benefit different cell types in different ways. The combination of cell-penetrating and non-penetrating CPAs also go hand in hand with the cryopreservation protocol, requiring careful considerations about the rates of CPA addition and removal. When penetrating CPAs are used, the risk of CPA toxicities is higher. Their removal from the cell prior to infusion also necessitates adjustments to the cryopreservation procedure. While DMSO is able to diffuse out of the cell, and equilibration of CPA removal must avoid too rapid a decrease in extracellular CPA concentration. For larger CPAs, the rates at which they will diffuse out of the cell are expected to be even slower. Cryopreservation approaches that rely on the introduction of polymers into cells, such as those utilizing nanoparticles, require additional considerations. Biodegradability, CPA release and removal and toxicity all need to be accounted for when these compounds and structures are expected to remain inside the cell after thawing.

Despite the progress, however, approaches which have resulted in robust solutions for the cryopreservation of immune cells are still in their infancy. As a result of the complexity of supplying adequate concentrations of CPA to immune cells while minimizing cryoinjury during freezing and thawing, and despite success with some tissues [142–144] and cell types—such as MSCs [100,145], red blood cells [146,147], fibroblasts [101] or induced pluripotent stem cells (iPSCs) [86,148]—very few of these attempts have been shown effective on T or NK cells.

4.4 Lyophilization as an alternative to cryopreservation

As an alternative to cryopreservation, lyophilization is of interest, but has so far only been attempted with bacterial cells[149], erythrocytes and platelets [150,151], umbilical cord blood-derived mononuclear cells [152], mesenchymal stem cells[153] and fibroblasts[154], largely in the presence of lyo-stabilizers such as trehalose. However, many of these studies have reported significant hemolysis and loss of viability. [154], [155]. Replacing trehalose or combining it with late embryogenesis abundant (LEA) proteins could further improve lyophilization by minimizing dehydration-induced stresses [156–158]. Despite this and the known issues with freezing immune cells, attempts at freeze-drying immune cells, such as primary T or NK cells or their engineered counterparts, do not yet exist. The belief that a viable strategy for the lyophilization of immune cells exists is part of the hope that the freeze-drying of these cells will one day become reality.

5. Cryopreservation of NK cells

Cryopreservation has been recognized as a major challenge to the use of NK cells in cancer immunotherapy [159]. Perhaps more so than many other immune cell types, NK cells have not responded favorably to cryopreservation, although post-thaw treatment conditions have been able to mitigate these issues somewhat. One belief is that NK cells are fragile and thus particularly sensitive to cryopreservation stresses, which they have a hard time surviving. Numerous studies have shown that the presence of DMSO leads to impaired cytolytic capacity of NK cells, and has been associated with the appearance of a phenotypically distinct, functionally-compromised CD56^{dim}CD16⁺ subset[160]. Cryopreservation with DMSO was also shown to lead to a reduced expression of tumor necrosis factor (TNF)-related apoptosis-inducing ligand (TRAIL) and natural killer group 2 D (NKG2D) ligands, alongside impaired NK cell cytotoxicity, which could be rescued by stimulation with IL-2.[161] The importance of IL-2 in sustaining NK cell effector function following cryopreservation was corroborated by other studies [162].

Immediate post-thaw viability of NK cells is typically higher than after a period of culture, but these cells have lower cytotoxic activity, including antibody-dependent cellular cytotoxicity and degranulation capacity, than post-thaw cultured cells [163]. NK cells cryopreserved in 10% DMSO, 40% Hank's balanced salt solution (HBSS) and 50%, of 25% human serum albumin (HSA), despite showing over 70% viability post-thaw, failed to lyse K562 targets unless rested overnight [164]. Additionally, these results were highly donor-variable. Miller *et al.* [165], studied *ex vivo* expanded or fresh cytokine-activated NK cells immediately after thawing or after overnight culture with IL-2. Generally, cells that had been cultured overnight showed superior cytotoxicity to cells infused immediately post-thaw. However, both groups of NK cells showed a drop in viability after overnight culture—*ex vivo* expanded NK cells had a viability of 20%, while the viability of freshly-activated NK cells was higher—73%—post-thaw and culture, but still lower than >90% prior to freezing. When both of these groups were infused into non-obese diabetic (NOD)-scid IL2 γ (null) (NSG) mice, only fresh NK cells that had been cryopreserved and cultured overnight expanded to a similar extent as fresh, non-cryopreserved NK cells, indicating that cryopreservation of *ex vivo*-expanded NK cells might impact their post-thaw functionality.

More recently, cryopreservation was reported to impair NK cell migration. The authors showed that the fraction of motile NK cells, evaluated using a 3D collagen gel, was 6-fold lower after cryopreservation compared to fresh cells. Cryopreserved NK cells also killed collagen gel-embedded tumor cells at a 5.6-fold lower rate compared to fresh cells [166].

On the other hand, studies have also shown that cryopreservation of NK cells is able to retain functionality and cytotoxic capacity of NK cells similar to those of fresh cells. In a study by Torelli *et al.* [167], cryopreserved NK cells (in HSA and DMSO) that had been expanded *ex vivo* in the presence of IL-2, IL-15 and feeder cells showed no functional impairment to the cells' cytotoxic capacity. Similarly, NK cells differentiated from umbilical cord blood CD34⁺ cells were able to be cryopreserved without loss of phenotype and function following thawing [168]. Exact reasons for the differences in observed responses to cryopreservation are unclear, however they could be attributable to expansion conditions (IL-2 alone *vs.* IL-2 and IL-15), source of cells (cord blood-differentiated *vs.* peripheral blood-derived NK cells), donor variability or simply freezing conditions. It is likely that a combination of factors has played a role.

Attempts at cryopreserving NK cells with media devoid of DMSO have begun to appear. In Table 2, we list all of the approaches that have so far been reported to utilize non-DMSO media for the cryopreservation of NK cells. As can be seen, these have so far been evaluated on NK-92 cells rather than primary human NK cells. We recently described the cryopreservation of the clinically-validated NK cell line NK-92 using a poly-L-lysine, dextran and ectoine-based CPA solution, which showed comparable NK cell viability and cytotoxicity against K562 targets as those obtained for DMSO-containing media [169]. Similarly, a combination of dextran-40 and carboxylated poly-L-lysine alone was also shown to result in potent retention of NK cell viability and cytotoxicity following cryopreservation that matched those of 10% DMSO-containing freezing media [170]. We also developed a nanoparticle-based approach for the intracellular delivery of trehalose to NK cells using chitosan-TPP nanoparticles. Using such nanoparticle-based system, we were able to recover NK cell viability post-thaw similarly to DMSO-containing cryopreservation media, while retaining superior cytotoxic functions against cancer targets [141].

However, cryopreservation of primary NK cells without DMSO remains difficult. These cells are challenging to expand and are prone to cryoinjury. Moreover, modulation of membrane permeability to shuttle cryoprotectants intracellularly or, generally, genetic manipulation of these cells might be difficult as NK cells display resistance to exogenous manipulation. This has been suggested to occur due to their pattern recognition receptors having evolved to trigger apoptosis mechanisms upon recognition of foreign material, such as genetic cargo [171]. Nonetheless, preliminary studies have shown that it may be possible to develop cryoprotective solutions for these cells that do not contain DMSO.

6. Cryopreservation of T cells

Adoptive transfer immunotherapy with T cells relies on various subsets of functionally and phenotypically distinct T cell subsets. These include, primarily, cytotoxic CD8⁺ T cells, helper CD4⁺ T cells, $\gamma\delta$ T cells, natural killer T (NKT) cells, and regulatory CD4⁺ T cells

(T_{regs}). All T cells are typically CD3⁺ and differ in expression of other surface proteins. CD8⁺ cytotoxic T cells are able to kill targets upon binding to peptides associated with major histocompatibility complex (MHC) class I molecules. These cells are most commonly used as effectors in adoptive transfer immunotherapy, including chimeric antigen receptor (CAR) therapy. CD4⁺ helper T cells are activated by antigens presented by MHC class II molecules. They are involved in priming of B cell responses and modulating CD8⁺ T cell immunity via secretion of cytokines, including interferon- γ (IFN- γ) and tumor necrosis factor- α (TNF- α). Adoptive transfer of CD4⁺ helper T cells was shown to enhance anti-tumor responses of CD8⁺ T cells [172]. $\gamma\delta$ T cells are considered to possess both adaptive- and innate-like features, and are able to recognize stress molecule on malignant cells irrespective of MHC haplotype, while avoiding graft-versus-host disease [173]. This makes them particularly attractive effectors in adoptive cellular therapy. NKT cells have features of both NK and T cells, and are CD3⁺CD56⁺. They recognize antigens presented by the MHC class I-like protein CD1d. T_{regs}, on the other hand, are highly immunosuppressive, and have, as such, found clinical relevance in adoptive transfer therapy for autoimmune diseases [174,175]. They express CD4, CD25 and FoxP3, and are CD127^{low}.

Each of these subsets responds differently to cryopreservation. The susceptibility of T cells to cryopreservation-induced damage has been long known [176]. Within the context of cryopreservation with DMSO, numerous studies have focused on the influence of freezing and thawing procedures on T cell viability, functionality and fitness [177].

T cells were shown, in a recent report by the Biomedical Excellence for Safer Transfusion Collaborative Group, to be more sensitive to cryopreservation than other cell types, such as HSCs [110], though not NK cells. For instance, T_{regs} have exhibited a reduction in viability, abnormal cytokine secretion and compromised surface marker expression due to DMSO [178,179]. Additionally, recent data [180] have suggested that compromised expression of surface markers on T_{regs} should mandate analysis be carried out on fresh samples to eliminate bias. To further complicate matters, T cells have significantly lower membrane permeability to CPAs and water than other cell types, such as oocytes, cancer cells and megakaryocytes [130], a characteristic that is even more pronounced in sick patients [181].

Overall, the scientific consensus on the effect of cryopreservation on T cell functions remains divergent. This is, in large part, due to the heterogeneous nature of the samples being sourced for analysis, including highly varied human pathologies from clinical studies. Some studies have shown a limited incidence of functional impairment, as measured by IFN- γ and cytokine production, of CD4⁺ and CD8⁺ T cells cryopreserved with DMSO [182]. Campbell *et al.* [183] reported a decreased expression of PD-1 and PD-L1 on PBMC-derived CD3⁺/CD8⁺ T cells following cryopreservation, hinting at potentially enhanced effector functions induced by the freezing and thawing process. More recently, Galeano Niño *et al.* [184] found that the cryopreservation process in the presence of 10% DMSO does not appreciably alter CD8⁺ T cells' cytotoxic function or ability to produce pro-inflammatory cytokines.

This is contrasted by data from other studies, which report significant functional impairment of T cell subsets following cryopreservation. Ford and colleagues [185], for instance,

reported a reduction in CD3⁺CD4⁺ T cells from frozen/thawed PBMC samples of patients following vaccination. Changes in T cell phenotype post-cryopreservation, such as alterations in levels of CD62L and CD45RA, have been also reported, alongside a loss of CD4⁺ and CD8⁺ naïve T cells [186,187]. Holthaus *et al.* [188] reported that the presence of as little as 0.25% DMSO in cell cultures could impair CD4⁺ T cell activation and signaling pathways. A higher concentration of 1% completely eliminated CD4⁺ T cell proliferation. Moreover, expression of markers CD69, CD25 and CD154 was impaired, IL-21⁺, IL-4⁺, and IL-22⁺ CD4⁺ T cells decreased in numbers, and glucose uptake was inhibited in the presence of DMSO. However, numbers of IFN- γ -producing CD4⁺ T cells were unaffected. Studies with T cell subsets sourced from patients have also explored the effects of cryopreservation. The ability of CD4⁺ and CD8⁺ T cells from acutely- and chronically-infected HIV individuals to produce IFN- γ after cryopreservation was shown to be affected by long-term cryopreservation [189]. IFN- γ production was impaired in CD4⁺ T cells in response to antigenic stimulation with proteins or individual HIV peptides (but not CMV pp65 peptide pools), and in CD8⁺ T cells in response to protein antigen stimulation. The study also noted that cryopreservation increased CD4⁺ T cell apoptosis after antigenic stimulation, with long term storage over 300 days leading to the most detrimental effect on cell function.

Generally, resting T cells after thawing helps their recovery, while culture conditions have been shown to affect their responses to cryopreservation. Luo *et al.* [190] reported that freezing CD3⁺CD4⁺ and CD3⁺CD8⁺ T cells at either day 8 or 12 during culture may improve their post-thaw function and recovery compared to cryopreservation at earlier time points. They also found that CD3⁺CD56⁺ NKT cell numbers were unchanged when these cells were harvested and frozen at culture day 12, but decreased at earlier time points. On the other hand, numbers of CD25⁺CD127⁻ T_{regs} were unchanged at 8 and 12 days, but increased in proportion at earlier time points. Sadeghi and colleagues [191] argued for a short recovery time of PBMC-derived CD3⁺ T cells post-thaw prior to infusion for maximum activity. They also observed a significant increase in expression of FoxP3 after culture of cryopreserved cells in IL-2. FoxP3 expression decreased in IL-2-free medium. In another study, Lemieux and colleagues [192] showed, using multiparametric flow cytometry, that a short period of rest (1 h)—but not longer incubation times—following thawing can aid in the recovery of some T cell phenotypes. Specifically, the proportion of CD4⁺CD8⁺ T cells was comparable to fresh samples after 1 h rest. They also noted higher loss of viability of memory T_{regs} (CD45RO⁺) than naïve T_{regs} (CD45RO⁻) immediately after cryopreservation, though they were similar to fresh samples after 1 h rest. Interestingly, they observed a loss of CD3 expression immediately after the first freeze-thaw cycle, but no further loss after the second. Clinical implementation of a resting step following thawing of the T cell therapy product would require closed systems and strict regulatory controls at the bedside, however, further increasing cost and processing burden. Eliminating the need for a wash step, by replacing DMSO with directly-infusible solutions, would be preferred.

Encouraging data from clinical studies utilizing cryopreserved CAR-T cells in patient infusions have indicated that these cells are able to tolerate cryopreservation, and possess equivalent *in vivo* persistence to fresh CAR-T products[193]. Analysis of 158 cultures of CAR-T cells from 6 clinical studies showed that the post thaw viability of CAR T-cells when

expanded post-transduction was superior to that of cryopreserved non-engineered peripheral blood CD3⁺ cells. Similarly, CAR-T cells cryopreserved both pre- and post-transduction and expansion were able to meet clinical specifications [193,194], leading the authors to conclude that cryopreservation of CAR-T cells is a viable clinical strategy. However, most CAR-T cell products utilize DMSO, often at concentrations above 5%[195].

Cryopreserving T cells without DMSO is considered difficult. Examples of DMSO-free solutions for T cell cryopreservation that have so far been evaluated and described in the literature are listed in Table 2. Despite the desire to replace DMSO from T cell cryopreservation media, no viable solution currently exists. Traditional T cell cryopreservation media rely on the supplementation of 10% DMSO with human serum albumin (HSA), FBS, PlasmaLyte[®] or human AB serum and the addition of sugars and small molecules [196]. Apart from replacing DMSO from lymphocyte solutions, washing it off after thawing the cells—typically with cell culture media—is a more common procedure within clinical settings [110,197–199]. These procedures, however, add additional processing steps to the cell therapy product prior to infusion. The direct removal of DMSO has also been attempted mechanically using a diffusion-based microfluidic device [200]. Commercial systems to wash off DMSO in a controllable manner have also been developed—Cesca Therapeutics' Thermogenesis X-Wash[™] is one such system[201].

Studies focusing on the cryopreservation of T cells with single molecules or osmolyte solutions in place of DMSO have largely been carried out using T cell lines, such as Jurkat cells, and not primary human cells [202]. These studies have used mixtures of osmolytes including sucrose, glycerol and isoleucine[203] or combinations of trehalose, sucrose, glycerol, mannitol and creatine[204] as cryoprotectants in place of DMSO, with higher concentrations of glycerol providing the most beneficial cryopreservation performance. A recent comparative study evaluating the DMSO-free cryopreservation of T cell subsets isolated from PBMCs, namely helper CD4⁺ T cells, cytotoxic CD8⁺ T cells and NKT cells, indicated that these T cell subsets respond differently to cryopreservation [111], and that ice crystal formation is not the only mechanism of cryoinjury that these cells experience during cryopreservation. Although all cell types were able to recover comparably to cells cryopreserved in DMSO, helper T cells showed the highest post-thaw recovery across all cryomedia tested, and this was most optimal with a combination of trehalose, glycerol and isoleucine, owing to this mixture's lowest enthalpy of melting. Contrary to helper T cells which were recovered in larger numbers in DMSO-free media compared to Jurkat cells, CD8⁺ T-cells (CD3⁺CD8⁺) had lower post-thaw recoveries with DMSO-free CPAs compared to Jurkat cells.

7. Conclusions

Successful cryopreservation of T or NK cells has been facing hurdles due to the long term toxicity of high concentrations of CPAs necessary for the logistic and deployment of cell therapies globally. Moreover, cryopreservation of T and NK cells with traditional media that contain DMSO directly affects the cells' immunomodulatory properties and subset makeup. With the approval of the first five CAR-T therapies in the United States, the need for safer, DMSO-free cryopreservation alternatives is critical. Despite many years of research,

there is no viable commercial product for the cryopreservation of T or NK cells devoid of DMSO. DMSO being universally used as a cryoprotectants is a growing concern, since emerging cell and gene therapies represent a new generation of clinical products that require a clinical option for cryopreservation that is directly-infusible, safe and functional. The use of DMSO as a CPA has led to unwanted expression of NK/T cell markers, diminished immune cell functionality, immune reactions and even death in certain clinical settings. New cryopreservation modalities that address and avoid detrimental physicochemical effects of freezing and thawing on T and NK cells are urgently needed. The ideal CPA formulation and cryopreservation technique would be clinically feasible and amenable to manufacturing scale, directly infusible, while avoiding the need to rest or culture the cells prior to infusion and removing the inconvenient, cost-sensitive, unsafe need to wash cells prior to treatment and delivery to medical centers of the released product.

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Key Points

- Infusions of cellular therapy products that contain DMSO are associated with significant toxicities in patients, requiring it to often be washed from the cellular product before infusion.
- Despite active research into alternatives to DMSO, the DMSO-free cryopreservation of NK and T cells is still in its infancy and remains extremely difficult: these cells are heterogeneous and very sensitive to cryopreservation.
- New technological approaches and a favorable regulatory landscape, especially in the context of CAR-T and CAR-NK cells, are fueling advances in combining delivery technologies with new formulations of bioinspired cryoprotectants that can achieve safe, infusible cryopreservation of NK and T cells.

Table 1.
Cryoprotective agents in DMSO-free cryopreservation.

CPA	Mode of action	Cell-penetrating/non-penetrating
α-tocopherol	ROS scavenger; Protection from oxidative stress	Non-penetrating
Albumin	Cell membrane stabilization	Non-penetrating
Acetamide	Inhibition of ice crystal formation	Cell-penetrating
Ammonium acetate	Maintenance of cellular ATP	Cell-penetrating
Trehalose	Membrane stabilization; water replacement	Non-penetrating
Dextran	Extracellular ice crystal protection	Non-penetrating
Glycerol	Intracellular cell protection; control of ice crystal growth	Cell-penetrating
Glucose	Membrane stabilization	Non-penetrating
Sucrose	Membrane stabilization	Non-penetrating
Trimethylamine oxide	Water attraction; maintenance of osmotic balance; inhibition of ice crystal growth	Cell-penetrating
Propylene glycol	Colligative cell protection	Cell-penetrating
Methanol	Water bonding; inhibition of ice crystal growth	Cell-penetrating
Ethylene glycol	Depression of freezing temperature; reduction in cell shrinkage; control of ice crystal growth	Cell-penetrating
Ectoine	Preferential exclusion	Non-penetrating
Formamide	Intracellular ice inhibition	Cell-penetrating
Polyvinylpyrrolidone	Colligative cell protection	Non-penetrating
Soybean lecithin	Membrane stabilization	Non-penetrating
Hydroxyethyl starch	Water absorption, glassy-state preservation	Non-penetrating
Poly-L-lysine	Ice crystal growth inhibition	Non-penetrating
Poloxamer 188	Cell membrane stabilization	Non-penetrating
Glycine	Maintenance of plasma membrane structural stability and fluidity	Non-penetrating
Butanediol	Hydrogen bonding with water; reduction in nucleation of ice	Cell-penetrating
Proline	Osmoprotection; ROS scavenging; protein stabilization	Non-penetrating
Isoleucine	Membrane stabilization	Non-penetrating
Creatine	Buffering cellular ATP	Non-penetrating
Mannitol	Osmotic stabilization	Non-penetrating
Xylose	Osmotic stabilization; decrease of intracellular water content	Non-penetrating
Pentaisomaltose	Osmotic stabilization	Non-penetrating

ATP: adenine triphosphate; ROS: reactive oxygen species

Table 2.

Examples of DMSO-free cryopreservation of NK and T cells.

Cell type	CPA	Post-thaw function
NK-92	Trehalose encapsulated in chitosan-TPP nanoparticles	Viability comparable to DMSO CPA; higher cytotoxicity compared to DMSO CPA[141]
NK-92	7.5% COOH-PLL + 5% dextran-40 + ectoine	Viability comparable to DMSO CPA[169]
NK-92	5% Dextran-40 + 7.5% COOH-PLL	Viability and cytotoxicity comparable to DMSO-based CPA[205]
Jurkat T cells	300 mM trehalose, 10% glycerol and 0.01% ectoine	Higher viability compared to DMSO CPA[202]
Jurkat T cells	Sucrose + glycerol + isoleucine	CPA optimization study; no direct comparison to DMSO CPA[203]
Jurkat T cells	Sucrose + glycerol; Trehalose + glycerol	Improved post-thaw function compared to DMSO[204]
PBMC-derived helper T cells; cytotoxic T cells; NKT cells	Sucrose + glycerol + isoleucine (SGI); Trehalose + glycerol + isoleucine (TGI); Maltose + glycerol + isoleucine (MGI)	Comparable recovery for helper and cytotoxic T cells to DMSO CPA; highest recovery with helper T cells in TGI; poor recovery for NKT cells[111]

COOH: carboxylated poly-L-lysine; CPA: cryoprotective agent; DMSO: dimethylsulfoxide; NK: natural killer